

The Foreign-Equity Strategy — A Plain-English Design

autoPortfolio: International Stocks (design / proposal, not yet built)

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What this document is

This is the blueprint for a third strategy: a foreign (international) stock portfolio to sit alongside the two you already run — US equities and FX carry. It is a design, not a deployed system. The goal here is to explain, in plain English, exactly how it would work, and — just as importantly — to be honest about whether it would actually help, because for one large chunk of the foreign universe the honest answer is probably “barely.”

Everything reuses the machinery you already trust. The interesting, new questions are only two: the currency and the diversification. Those get most of the space.

Part 1 — The core: same recipe as US equities

The foreign book is deliberately the same structure as your US equity book, because that structure already survived heavy testing (simple beat clever, every time). One sentence:

Hold an equal-weighted basket of the most-liquid foreign stocks, refreshed quarterly, diversified across sectors and regions, with the same 15% volatility-target safety overlay — and, on top, a currency decision.

The five pieces you already know, applied abroad:

1. Which stocks? The most liquid foreign names. Same as home: rank eligible foreign stocks by dollar trading volume, take the top N. Liquid names keep trading costs realistic — and abroad this matters more, because foreign markets are generally less liquid and more expensive to trade than the US.
2. How much of each? Equal weight ($1/N$). Same reason it won at home: $1/N$ is famously hard to beat out-of-sample. No change.
3. Diversified across sectors and regions (a cap on each). At home we cap ≤ 5 per sector. Abroad we add a second cap: $\leq$ some max per country/region, so the book can't quietly become an all-Japan or all-Europe bet. This is the one genuinely new selection rule.
4. The 15% volatility-target overlay. Identical: measure the book's recent volatility, move to cash when it spikes, stay near a steady 15%/yr. Never levers, only de-risks.
5. Point-in-time + null tests. Same discipline: at every past date use the names that were actually liquid then (no hindsight), test every design choice against a random-signal "null," validate on held-out periods. Nothing is trusted until it beats coin-flips.

If that were the whole story, this page would be the whole document. It isn't, because of the two foreign-specific problems below.

Part 2 — New problem #1: the currency (hedge or not)

A foreign stock is really two bets bundled together:

foreign-stock return (in USD) $\approx$ the local stock move + the currency's move vs the dollar

Buy a Japanese stock: even if it rises 10% in yen, a 10% fall in the yen wipes the gain when you convert back to dollars. So owning foreign stocks unhedged means you're also running a pile of currency bets — whether you meant to or not.

That's a problem for you specifically, because you already run a currency book (the carry sleeve). Stacking unmanaged FX exposure on top of a deliberately managed FX sleeve means you could be taking the same yen or peso risk twice, in sizes you never chose.

The fix — hedge the currency. Hedging means removing the FX portion (sell the foreign currency forward, same size as the position), leaving a clean bet on the foreign companies. The two sleeves then stay in their lanes: the equity sleeve bets on stocks, the carry sleeve bets on currencies, and neither accidentally doubles the other.

	Unhedged	Hedged (recommended)
You own	Stock + currency bet	Just the stock
Cost	Free	Small: the interest-rate gap between the two currencies
Overlaps carry book?	Yes — doubles FX risk	No — clean separation
Adds FX diversification?	Yes, but uncontrolled	No

Recommendation: hedge, at least for developed markets. It keeps the risk attributable to one place and makes the diversification question (next) clean — we can ask “do foreign companies diversify US companies?” without the answer being muddled by currency noise. One caveat: hedging costs roughly the carry differential, and for EM that cost is large — so for an EM sub-book we'd test hedged and unhedged and let the data decide.

Part 3 — New problem #2: does it even diversify? (the real question)

Running a third strategy is only worth it if it doesn't move with the first two. This is where foreign equities split into two very different animals.

Developed international (Europe, Japan, UK, Canada, Australia...)

The uncomfortable truth: developed international stocks are ~ 0.8 correlated with US stocks. They fall in the same crashes (2008, 2020, 2022) and rise in the same booms. They are almost the same bet in a slightly different wrapper.

This is the exact same situation as G10 carry, which the Markowitz optimizer gave 0% weight — not because it was bad, but because it was too similar to what you already held to earn a slice. Developed-international equity will very likely hit the same wall: the optimizer looks at it, sees “US stocks with extra steps,” and allocates near zero.

So we should expect developed international to earn a small-to-zero allocation, and we should be fine with that. Testing it is still worthwhile — but we go in expecting the answer “it's redundant,” not hoping otherwise.

Emerging-market equity (the interesting one)

EM equity (China, India, Brazil, Taiwan, Korea...) is the genuine diversifier candidate — it's only ~ 0.6 – 0.7 correlated with US stocks, driven by different growth stories, commodity cycles, and local politics. This is the direct analogue of widening carry to EM, which is precisely the move that worked (carry Sharpe $0.31 \rightarrow 0.51$, and it earned a real 29% Markowitz slice).

But EM equity carries its own crash tail — and here's the subtle trap: it may crash at the same time and for the same reason as your EM carry book. When the Mexican peso devalues, Mexican stocks usually fall too. So EM equity and EM carry could share a hidden common tail — the diversification looks great in calm data and then both lose together in exactly the crisis you were hedging.

This is the one thing we must measure before committing a dollar.

The gate it has to pass (before we build)

Same diversification threshold we applied to the carry sleeve: an asset earns a slice only if its risk-adjusted return clears the bar set by how correlated it is to what you already own. Concretely, before building we run:

1. Correlation check — foreign book vs US equity and vs the carry book, overall and in each sleeve's worst months (the tail correlation is what matters, not the average).
2. Markowitz with three assets — US equity, carry, foreign — and see what weight foreign actually earns. If it's ~ 0 (likely for developed), we learn that cheaply, before writing the engine.
3. Shared-tail check for EM — does EM equity crash with EM carry? If yes, the “diversification” is partly an illusion and we size it down hard.

Part 4 — What the finished strategy would look like

Putting it together, the deployed foreign book (if it passes the gate):

Piece	Choice	Same as US?
Universe	Most-liquid foreign stocks, split developed vs EM	new: two regions
Selection	Top-N by dollar volume, point-in-time	same
Weighting	Equal weight (1/N)	same
Diversification cap	$\leq M$ per sector and $\leq K$ per country/region	new: region cap
Rebalance	Quarterly	same
Risk overlay	15% volatility target (de-risk to cash)	same
Currency	Hedged (developed); test both for EM	new
Validation	Point-in-time, null tests, held-out periods	same

Honest expected verdict (before any data)

- Developed international, hedged: clean, safe, well-diversified internally — but ~0.8 correlated to US, so expect a near-zero portfolio slice. Probably not worth deploying on its own merits; it's "more of the same."
- EM equity: the real prize if it survives the shared-tail test with EM carry. If its crashes are independent enough, it earns a genuine slice like EM carry did. If they crash together, it's a smaller diversifier than it looks and gets sized down. This is the piece worth building — pending the test.
- Overall: the value of this strategy is concentrated in EM equity, and even there it's gated on one specific measurement. So the right first step is not to build the engine — it's to run the three-asset correlation/Markowitz check with a cheap proxy (e.g. regional index returns) and confirm foreign equity clears the bar. Build only what the diversification math says is worth building.

One-page summary

Question	Answer
Core recipe?	Same as US equity: liquid, equal-weight, sector-capped, quarterly, 15% vol-target, point-in-time
What's new?	(1) a currency hedge decision, (2) a region cap, (3) a developed-vs-EM split
Hedge the currency?	Yes for developed (keeps FX risk in the carry sleeve, not doubled); test both for EM

Question	Answer
Does developed intl diversify?	Barely — ~0.8 correlated to US, likely ~0% optimal weight (like G10 carry)
Does EM equity diversify?	Potentially yes (~0.6 correlated) — the real prize — but may share a crash tail with EM carry
What must we check first?	3-asset Markowitz + tail-correlation, especially EM-equity vs EM-carry, before building
Expected bottom line	Value is concentrated in EM equity, gated on the shared-tail test; developed intl is likely redundant

This is a design document. No foreign-equity engine has been built yet. The recommended first action is the diversification gate (Part 3) using cheap index proxies — build the full point-in-time engine only for the pieces that clear it.